

Translated

From: e-maxx.ru

Ternary Search

We are given a function $f(x)$ which is unimodal on an interval $[l, r]$. By unimodal function, we mean one of two behaviors of the function:

1. The function strictly increases first, reaches a maximum (at a single point or over an interval), and then strictly decreases.
2. The function strictly decreases first, reaches a minimum, and then strictly increases.

In this article, we will assume the first scenario. The second scenario is completely symmetrical to the first.

The task is to find the maximum of function $f(x)$ on the interval $[l, r]$.

Algorithm

Consider any 2 points m_1 , and m_2 in this interval: $l < m_1 < m_2 < r$. We evaluate the function at m_1 and m_2 , i.e. find the values of $f(m_1)$ and $f(m_2)$. Now, we get one of three options:

- $f(m_1) < f(m_2)$

The desired maximum can not be located on the left side of m_1 , i.e. on the interval $[l, m_1]$, since either both points m_1 and m_2 or just m_1 belong to the area where the function increases. In either case, this means that we have to search for the maximum in the segment $[m_1, r]$.

- $f(m_1) > f(m_2)$

This situation is symmetrical to the previous one: the maximum can not be located on the right side of m_2 , i.e. on the interval $[m_2, r]$, and the search space is reduced to the segment $[l, m_2]$.

- $f(m_1) = f(m_2)$

We can see that either both of these points belong to the area where the value of the function is maximized, or m_1 is in the area of increasing values and m_2 is in the area of descending values (here we used the strictness of function increasing/decreasing).